

Original practice 4: Experimental decisions and design challenges

Original prediction practice—not official questions or a guarantee

Total: 36 marks • Suggested time: 75 minutes

Fair tests, wire geometry, current limits for a low-voltage design, a branched series circuit and evidence-limited conclusions.

Status and instructions

Original prediction practice—not official questions or a guarantee

Author-created chapter practice, not a claim about an official examination format. All 16 questions are compulsory. Show working and units. Circuit wires, meters and sources are ideal unless stated; resistor temperatures remain constant unless a question explicitly studies a change. Treat stated data as exact unless a tolerance is given.

All case datasets and SVGs are original constructed teaching material, not measurements reported by NCERT. NCERT is cited for concepts and formulas, not as the source of these questions.

- A: Q1-6, $6 \times 1 = 6$ marks
- B: Q7-10, $4 \times 2 = 8$ marks
- C: Q11-13, $3 \times 3 = 9$ marks
- D: Q14, $1 \times 5 = 5$ marks
- E: Q15-16, $2 \times 4 = 8$ marks
- Total: 16 questions, 36 marks

Official curriculum scope and assessment limitation

CBSE Science (086), Class X, 2026-27, PDF p.6 Unit IV lists electric current, potential difference, resistance, resistivity, Ohm's law, resistor combinations, heating and power. Practical 4 on PDF p.7 covers V-I measurements; practical 5 on PDF p.8 covers series and parallel equivalent resistance. Source: https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Science_SecP1_2026-27.pdf

The same PDF p.6 contains an unresolved teacher-note wording inconsistency naming “Electric Effects of Electric Current” among unassessed topics, despite the explicit Unit IV list. These are chapter study materials, not a guarantee of summative assessment coverage; consult school/CBSE clarification. Unit IV's 13 marks covers a combined unit, not an Electricity-only allotment.

Section A — MCQ

1. [1 marks]

1. Two uniform cylindrical wire samples have the same material, length and temperature. Sample B has twice sample A's diameter. What is R_B/R_A ?

- A. 2
- B. $1/2$
- C. 4
- D. $1/4$

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2. [1 marks]

2. Which change is appropriate for testing only the effect of length on wire resistance?

- A. Change both material and length
- B. Change length but keep material, area and temperature fixed
- C. Change length and diameter together
- D. Change temperature and length together

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3. [1 marks]

3. A uniform wire has resistance $16\ \Omega$. It is divided into two equal lengths without changing its cross-sectional area, material or temperature. What is the resistance of one piece?

- A. $8\ \Omega$
- B. $32\ \Omega$
- C. $4\ \Omega$
- D. $16\ \Omega$

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4. [1 marks]

4. Which ordered pair of units correctly represents resistance and resistivity?

- A. $\Omega\ \text{m}$, Ω
- B. A, V
- C. Ω , $\Omega\ \text{m}$
- D. Ω/m , Ω

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5. [1 marks]

5. A low-voltage test resistor dissipates 7.2 W when the current is 0.60 A. What voltage is across it?

- A. 4.32 V
- B. 0.083 V
- C. 20 V
- D. 12 V

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.8, printed pp191–192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; 1 kWh = 3.6×10^6 J. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

6. [1 marks]

6. A V-versus-I graph has voltage vertical and current horizontal. Line P is steeper than line Q. Both are straight through the origin on the same scales. Which conclusion follows?

- A. P has lower resistance
- B. P has higher resistance
- C. The resistances are equal
- D. P must have a larger current at every voltage

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.4, printed pp175–177; Activity 11.1 p175; Eqs. (11.4)–(11.7) p176: $V = IR$ at constant temperature; Activity 11.2 pp176–177 compares components. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

Section B — 2-mark questions

7. [2 marks]

7. At fixed temperature, an initially uniform wire is changed to a uniform wire of twice its original length while its volume and material remain unchanged. Find the new area in terms of the old area and the resistance ratio.

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8. [2 marks]

8. A 15Ω test component must operate at 0.20 A. Calculate the required voltage and its power.

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9. [2 marks]

9. A 10 V low-voltage supply delivers 0.40 A to a sealed resistor box. Find its equivalent resistance. Explain why this reading does not reveal how many resistors are inside.

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10. [2 marks]

10. A student labels 2 kW as an amount of energy and 7200 J as a power. Correct both labels and calculate the time for a 2 kW device to transfer 7200 J.

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Section C — 3-mark questions

11. [3 marks]

11. Two uniform wires have equal length, cross-sectional area and temperature, but different materials. At 3 V, wire U carries 0.60 A and wire V carries 0.15 A. Find (a) R_U , (b) R_V and (c) ρ_V/ρ_U .

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12. [3 marks]

12. A 6 V model uses two identical 12Ω resistors. It needs total current no greater than 0.60 A. Test (a) series current and (b) parallel current, then (c) choose the arrangement that satisfies the stated limit. Use both resistors.

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13. [3 marks]

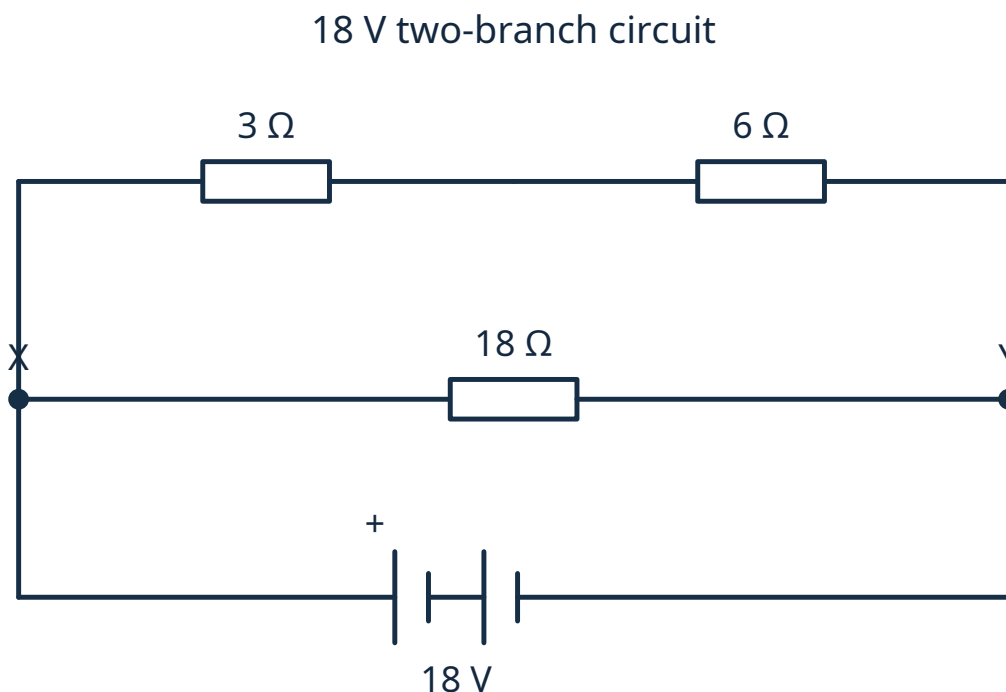
13. Design a fair comparison of currents through two different components at the same voltage, using an ammeter and a voltmeter. State (a) meter connections, (b) what to hold the same, and (c) how the readings reveal which component has larger operating resistance.

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Section D — 5-mark questions

14. [5 marks]

14. In the supplied 18 V circuit, the upper branch has $3\ \Omega$ and $6\ \Omega$ in series; the lower branch has $18\ \Omega$. Find (a) upper-branch resistance, (b) current in each branch (1 mark together), (c) total current, (d) equivalent resistance and (e) voltage across the upper $6\ \Omega$ resistor. Each part is worth 1 mark.



Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.6.1, printed pp182–185; Activity 11.4 p182 (same current); Activity 11.5 p183 (voltage sum); Eqs. (11.11)–(11.14) pp183–184. §11.6.2, printed pp185–187; Activity 11.6 pp185–186; Eqs. (11.15)–(11.18) p186: branch-current sum and reciprocal-resistance sum. §11.4, printed pp175–177; Activity 11.1 p175; Eqs. (11.4)–(11.7) p176: $V = IR$ at constant temperature; Activity 11.2 pp176–177 compares components. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

Section E — Case-based questions

15. [4 marks]

15. Constructed fair-test records, all with the same material and temperature at 2 V: wire A has length l and area S , current 0.40 A; B has length $2l$ and area S , current 0.20 A; C has length l and area $2S$, current 0.80 A. (a) Find all three resistances (1). (b) Identify the pair testing length alone and its conclusion (1). (c) Identify the pair testing area alone and its conclusion (1). (d) Explain why comparing only B and C does not isolate either factor (1).

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.5, printed pp177–180; Activity 11.3 pp177–178; Eqs. (11.8)–(11.10) p178: $R = \rho l/A$; material and temperature discussion p179. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

16. [4 marks]

16. Constructed component readings: at 1, 2 and 3 V, component X carries 0.10, 0.20 and 0.30 A; component Y carries 0.10, 0.16 and 0.20 A. Both begin at the same temperature, but temperature is not controlled during a run. (a) Find X's V/I values (1). (b) Find Y's V/I values (1). (c) State what the data support without assigning a proven cause (1). (d) Specify a follow-up test that would investigate heating as an explanation (1).

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Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — original-practice-04-q01

[1 mark] D. Area is proportional to diameter squared, so B has four times the area and $R_B/R_A = 1/4$.

Question 2 — original-practice-04-q02

[1 mark] B. Keep material, area and temperature unchanged and vary length.

Question 3 — original-practice-04-q03

[1 mark] A. At unchanged material, area and temperature, $R \propto l$; each half has $16/2 = 8 \Omega$.

Question 4 — original-practice-04-q04

[1 mark] C. Resistance is in Ω ; from $\rho = RA/l$, resistivity is in $\Omega \text{ m}$.

Question 5 — original-practice-04-q05

[1 mark] D. $V = P/I = 7.2/0.60 = 12 \text{ V}$.

Question 6 — original-practice-04-q06

[1 mark] B. Here slope = $\Delta V/\Delta I = R$, so $R_P > R_Q$.

Question 7 — original-practice-04-q07

[1 mark] Volume IA is fixed, so $2I \times A_{\text{new}} = IA$; $A_{\text{new}} = A/2$.

[1 mark] $R_{\text{new}}/R_{\text{old}} = (2I/(A/2))/(I/A) = 4$; resistance quadruples.

Question 8 — original-practice-04-q08

[1 mark] $V = IR = 0.20 \times 15 = 3 \text{ V}$.

[1 mark] $P = VI = 3 \times 0.20 = 0.60 \text{ W}$.

Question 9 — original-practice-04-q09

[1 mark] $R_{\text{eq}} = V/I = 10/0.40 = 25 \Omega$.

[1 mark] One 25Ω resistor or many combinations can give 25Ω , so a terminal V/I measurement does not uniquely determine component count.

Question 10 — original-practice-04-q10

[1 mark] 2 kW is power; 7200 J is energy (0.5 mark for each correction).

[1 mark] $t = E/P = 7200/2000 = 3.6 \text{ s}$.

Question 11 — original-practice-04-q11

[1 mark] (a) $R_U = 3/0.60 = 5 \Omega$.

[1 mark] (b) $R_V = 3/0.15 = 20 \Omega$.

[1 mark] (c) Equal I and A give $\rho_V/\rho_U = R_V/R_U = 20/5 = 4$.

Question 12 — original-practice-04-q12

[1 mark] (a) Series: $R = 24 \, \Omega$; $I = 6/24 = 0.25 \, \text{A}$.

[1 mark] (b) Parallel: $R = 6 \, \Omega$; $I = 6/6 = 1.0 \, \text{A}$.

[1 mark] (c) Choose series: $0.25 \, \text{A} \leq 0.60 \, \text{A}$; parallel exceeds the specified design limit.

Question 13 — original-practice-04-q13

[1 mark] (a) Put the ammeter in series with the test component and the voltmeter across it.

[1 mark] (b) Set the same measured voltage across each component and maintain specified equal test-temperature conditions; replace only the component.

[1 mark] (c) Calculate $R = V/I$ for each. At equal V , the smaller current indicates the larger operating resistance; this does not alone prove constant R at other voltages.

Question 14 — original-practice-04-q14

[1 mark] (a) $R_{\text{upper}} = 3 + 6 = 9 \, \Omega$.

[1 mark] (b) $I_{\text{upper}} = 18/9 = 2 \, \text{A}$; $I_{\text{lower}} = 18/18 = 1 \, \text{A}$ (0.5 mark each).

[1 mark] (c) $I_{\text{total}} = 2 + 1 = 3 \, \text{A}$.

[1 mark] (d) $R_{\text{eq}} = V/I_{\text{total}} = 18/3 = 6 \, \Omega$.

[1 mark] (e) $V_6 = I_{\text{upper}} \times 6 = 2 \times 6 = 12 \, \text{V}$; it is not the full 18 V because the upper resistors share the branch voltage.

Question 15 — original-practice-04-q15

[1 mark] (a) $R_A = 2/0.40 = 5 \, \Omega$; $R_B = 2/0.20 = 10 \, \Omega$; $R_C = 2/0.80 = 2.5 \, \Omega$ (award 1/3 mark each).

[1 mark] (b) A and B: doubling length at fixed area doubles resistance.

[1 mark] (c) A and C: doubling area at fixed length halves resistance.

[1 mark] (d) Both length and area change between B and C, so their separate effects cannot be isolated by that comparison alone.

Question 16 — original-practice-04-q16

[1 mark] (a) X: $1/0.10 = 2/0.20 = 3/0.30 = 10 \, \Omega$.

[1 mark] (b) Y: 10 Ω , 12.5 Ω and 15 Ω .

[1 mark] (c) X has constant V/I over these readings; Y does not. Heating is a possible explanation for Y, but these readings do not establish it.

[1 mark] (d) Repeat brief measurements while controlling or monitoring component temperature and cooling to the same baseline; compare V/I at the same temperatures to test the heating explanation.